1 Funding Analysis and Grant-to-Publication Linkage

1.1 Learning objectives

After completing this chapter, you will be able to:

- Extract funder information from OpenAlex work records
- Compute funder-level publication counts and citation impact
- Analyse co-funding patterns (papers acknowledging multiple funders)
- Interpret funding data critically, accounting for coverage and attribution challenges
- Discuss why funding-output linkage is inherently imprecise

1.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "funding_helpers.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Funding agencies invest billions in research and want to understand what their investment produces. **Funding acknowledgement analysis** extracts funder names and grant identifiers from publication metadata or full text, linking grants to the papers they supported.

OpenAlex includes funder information for many works, derived from Crossref's funding metadata (which in turn comes from publisher-reported FundRef data). Coverage varies: journals that participate in Crossref's FundRef initiative have good coverage; others may have no funding data at all [@priem2022].

Key analytical questions include: How many publications does a funder support? What is the citation impact of funded vs. unfunded research? Which funders frequently co-fund research? How does a funder's portfolio map onto the landscape of science?

The fundamental challenge is **attribution**: a paper acknowledging three funders and seven authors is the result of contributions from all of them. Assigning full credit to each funder double-counts; fractional credit is more principled but harder to implement. Similar issues arise with multi-source funding: a paper may acknowledge an infrastructure grant, a project grant, and a fellowship, each with different levels of contribution.

1.4 Worked example

1.4.1 Extracting funder data

```
works <- fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 400, seed = 42)
)

funding_works <- works |>
  mutate(funders = map(funders, sci_funder_rows),
         has_funder_metadata = map_int(funders, nrow) > 0)

funded <- funding_works |>
  filter(has_funder_metadata) |>
  select(id, display_name, cited_by_count, funders)

cat(glue("Works with observed funder IDs: {nrow(funded)} / {nrow(works)}
 ↪  ({scales::percent(nrow(funded)/nrow(works))})\n"))
```

```
#> Works with observed funder IDs: 220 / 400 (55%)
```

```
if (!nrow(funded)) {
  cat("No usable funder records occur in this sample. Funder summaries are unavailable;
   ↪  missing metadata does not establish that the papers were unfunded.\n")
}
```

```
funder_data <- if (nrow(funded)) {
  funded |>
    unnest(funders, names_sep = "_") |>
    select(work_id = id, funder_id = funders_id,
           funder = funders_display_name, cited_by_count)
} else {
  tibble(work_id = character(), funder_id = character(),
         funder = character(), cited_by_count = numeric())
}

cat(glue("Funder-work pairs: {nrow(funder_data)}\n"))
```

```
#> Funder-work pairs: 355
```

```
cat(glue("Unique funders: {n_distinct(funder_data$funder_id)}\n"))
```

```
#> Unique funders: 185
```

1.4.2 Top funders by publication count

```
funder_counts <- funder_data |>
  group_by(funder_id) |>
  summarise(funder = first(funder), n = n_distinct(work_id), .groups = "drop") |>
  arrange(desc(n))

if (nrow(funder_counts)) {
funder_counts |>
  head(15) |>
  mutate(funder = fct_reorder(funder, n)) |>
  ggplot(aes(x = n, y = funder)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Publications supported", y = NULL) +
  theme_sci()
} else {
  cat("No funder publication-count plot: no observed funder records.\n")
}
```

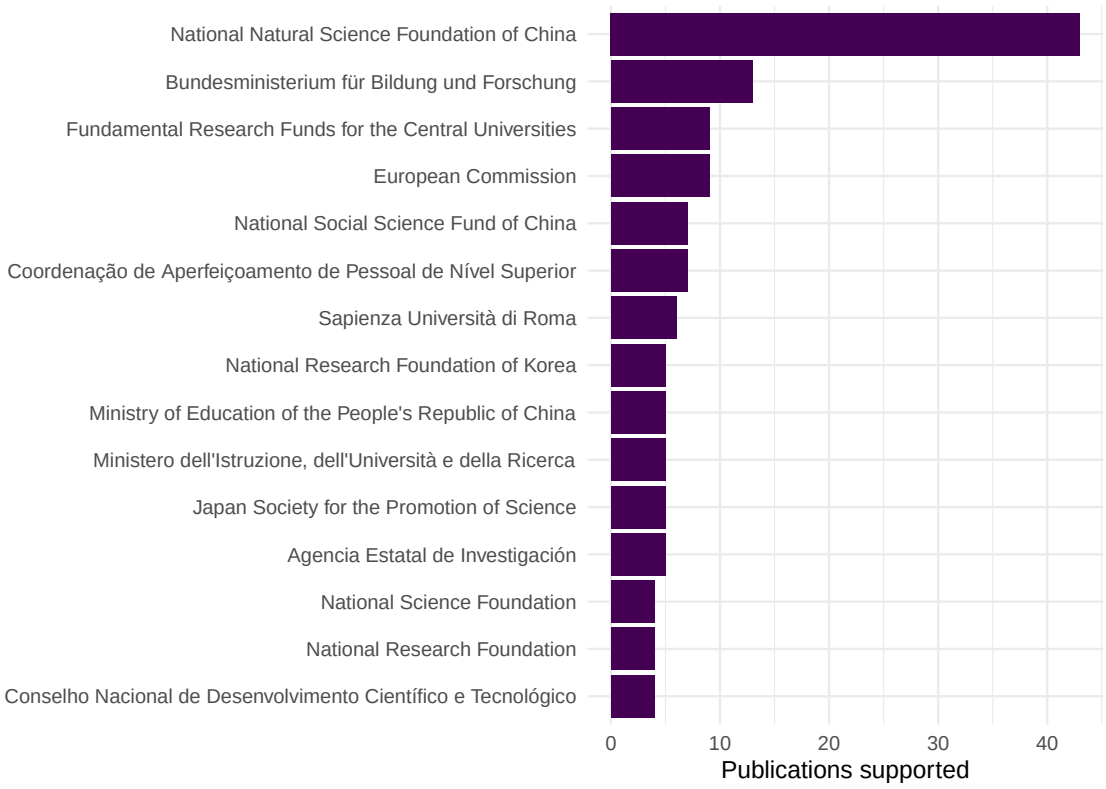


Figure 1: Up to 15 funders by the number of sampled publications with observed funder metadata.

1.4.3 Citation impact by funder

```
funder_impact <- funder_data |>
  group_by(funder_id) |>
  summarise(
    funder = first(funder),
```

funder_id	funder
https://openalex.org/F4320335869	National Social Science Fund of China
https://openalex.org/F4320335598	Agencia Estatal de Investigación
https://openalex.org/F4320322120	National Research Foundation of Korea
https://openalex.org/F4320321001	National Natural Science Foundation of China
https://openalex.org/F4320321091	Coordenação de Aperfeiçoamento de Pessoal de Nível Superi
https://openalex.org/F4320334764	Japan Society for the Promotion of Science
https://openalex.org/F4320321106	Ministry of Education of the People's Republic of China
https://openalex.org/F4320335787	Fundamental Research Funds for the Central Universities
https://openalex.org/F4320321114	Bundesministerium für Bildung und Forschung
https://openalex.org/F4320320300	European Commission

```
    n_pubs = n(),
    mean_cites = round(mean(cited_by_count), 1),
    median_cites = median(cited_by_count),
    .groups = "drop"
  ) |>
  filter(n_pubs >= 5) |>
  arrange(desc(mean_cites))

if (nrow(funder_impact)) {
  funder_impact |> head(10) |> gt()
} else {
  cat("No funder has at least five sampled publications with usable metadata; no funder
   ↪  citation summary is reported.\n")
}
```

1.4.4 Comparison by observed funder metadata

```
if (n_distinct(funding_works$has_funder_metadata) == 2L) {
funding_works |>
  mutate(funder_metadata = if_else(has_funder_metadata,
                                   "Observed funder", "No observed funder")) |>
  ggplot(aes(x = funder_metadata, y = cited_by_count + 1)) +
  geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
  scale_y_log10() +
  labs(x = "Funder metadata", y = "Citations (log scale)") +
  theme_sci()
} else {
  cat("No two-group citation comparison: only one funder-metadata group is observed.\n")
}
```

1.5 Diagnostics and interpretation

- **Coverage rate**: The fraction of papers with funder data depends on the journal, publisher, and field. Report this rate before interpreting funder statistics.

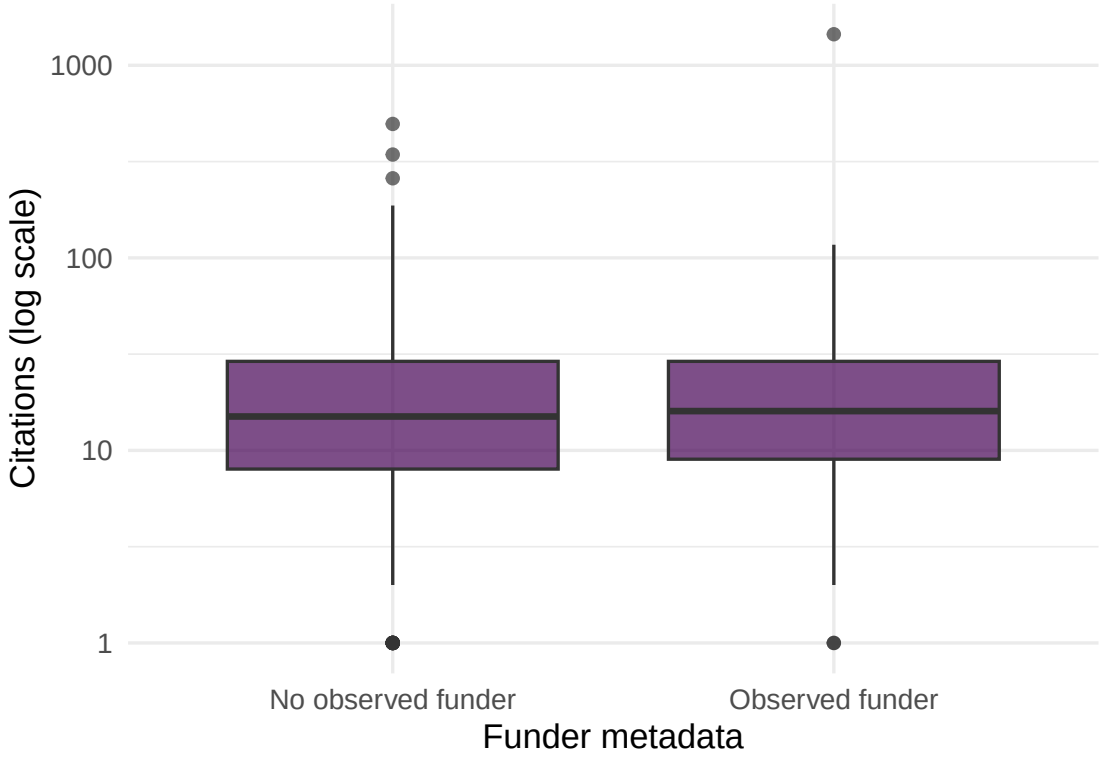


Figure 2: Citation distribution by presence of observed funder metadata; missing metadata does not establish absence of funding.

- **Funder name normalisation**: The same funder may appear under different names ("NSF", "National Science Foundation", "US NSF"). Merge variants using funder IDs where available.
- **Selection and metadata coverage**: Papers with observed funder records may differ systematically from papers without them. Their citation gap can reflect selection and publisher reporting; it does not identify the causal effect of funding or compare confirmed funded and unfunded groups.
- **Multi-funder attribution**: Papers acknowledging multiple funders inflate the publication count for each. Consider fractional counting.

1.6 Limitations and responsible use

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- **Coverage is incomplete.** Not all publishers report funding data to Crossref. Missing funder acknowledgements do not mean research was unfunded.
- **Attribution is ambiguous.** A grant acknowledgement does not specify which part of the work the grant supported. Infrastructure grants, salary support, and project funding all appear identically.
- **Correlation ≠ causation.** Funded papers receive more citations, but this may reflect that better-resourced labs produce both more funding applications and more cited papers.
- **Do not rank funders by citation impact.** Funders support different fields, career stages, and risk levels. Comparing mean citations across funders is misleading without field normalisation [@hicks2015].

1.8 Common pitfalls

1.9 Common pitfalls

- **Ignoring funder coverage rates.** Comparing funder portfolios without accounting for differential coverage produces biased results.
- **Treating every acknowledgement as equal.** A paper may acknowledge a major project grant and a minor travel grant. Both appear identically in metadata.
- **Comparing across fields.** A biomedical funder's publications will have higher citation counts than a humanities funder's publications due to field-level citation norms.
- **Using funder data for individual evaluation.** Whether a researcher has funding reflects many factors beyond research quality: field norms, career stage, institutional support, and luck.

1.10 Exercises

1. **Funder co-occurrence.** Identify papers with multiple funders. Which funder pairs most frequently co-fund research?
2. **Funder portfolios.** For two major funders, fetch their supported publications and compare topical profiles using keyword analysis.
3. **Temporal trends.** Track the number of papers with funding acknowledgements by year. Is the coverage rate increasing over time?

1.11 Solutions

Solutions are provided in **??**.

1.12 Further reading

- @priem2022 — OpenAlex funder metadata from Crossref FundRef.
- @hicks2015 — Responsible use of metrics in funding evaluation.
- @waltman2016 — Citation indicators and their application to funding assessment.

1.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
 ↪  version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.5                Matrix_1.7-5
#>  [3] word2vec_0.4.1            stm_1.3.8
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2
#>  [7] visNetwork_2.1.4          ggraph_2.2.2
#>  [9] tidygraph_1.3.1           igraph_2.3.4
#> [11] quanteda_4.5.1            pdftools_3.9.1
#> [13] arrow_26.0.0              bibliometrix_5.5.0
#> [15] RefManageR_1.4.0          bib2df_1.1.2.0
#> [17] rcrossref_1.2.1           gt_1.4.0
#> [19] tidytext_0.4.3            glue_1.8.1
#> [21] openalexR_3.1.2           lubridate_1.9.5
#> [23] forcats_1.0.1             stringr_1.6.0
#> [25] dplyr_1.2.1               purrr_1.2.2
#> [27] readr_2.2.0               tidyr_1.3.2
#> [29] tibble_3.3.1              ggplot2_4.0.3
#> [31] tidyverse_2.0.0           bookdown_0.48
#> [33] fs_2.1.0                  rmarkdown_2.32
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.3           RColorBrewer_1.1-3     jsonlite_2.0.0
#>   [4] magrittr_2.0.5         modeltools_0.2-25      farver_2.1.2
#>   [7] vctrs_0.7.3            memoise_2.0.1          askpass_1.2.1
#>  [10] base64enc_0.1-6        tinytex_0.61           htmltools_0.5.9
#>  [13] contentanalysis_1.1.1  curl_8.1.0             janeaustenr_1.0.0
```

```
#>  [16] cellranger_1.1.0       htmlwidgets_1.6.4      tokenizers_0.3.0
#>  [19] plyr_1.8.9             httr2_1.3.0            plotly_4.12.1
#>  [22] cachem_1.1.0           dimensionsR_0.0.3      mime_0.13
#>  [25] lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [28] fastmap_1.2.0          shiny_1.14.0           digest_0.6.39
#>  [31] patchwork_1.3.2        shinycssloaders_1.1.0  rprojroot_2.1.1
#>  [34] RSpectra_0.16-2        SnowballC_0.7.1        labeling_0.4.3
#>  [37] urltools_1.7.3.1       timechange_0.4.0       mgcv_1.9-4
#>  [40] httr_1.4.9             polyclip_1.10-7        compiler_4.6.1
#>  [43] here_1.0.2             bit64_4.8.6            withr_3.0.3
#>  [46] S7_0.2.2               viridis_0.6.5          ggforce_0.5.0
#>  [49] MASS_7.3-65            bibliometrixData_0.3.0 tools_4.6.1
#>  [52] otel_0.2.0             stopwords_2.3          zip_3.0.2
#>  [55] rentrez_1.2.4          httpuv_1.6.17          nlme_3.1-169
#>  [58] promises_1.5.0         grid_4.6.1             stringdist_0.9.17
#>  [61] reshape2_1.4.5         generics_0.1.4         gtable_0.3.6
#>  [64] tzdb_0.5.0             rscopus_0.9.0          ca_0.72
#>  [67] data.table_1.18.6.1    hms_1.1.4              xml2_1.6.0
#>  [70] utf8_1.2.6             ggrepel_0.9.8          pillar_1.11.1
#>  [73] nsyllable_1.0.1        vroom_1.7.1            later_1.4.8
#>  [76] splines_4.6.1          tweenr_2.0.3           lattice_0.22-9
#>  [79] FNN_1.1.4.1            bit_4.6.0              tidyselect_1.2.1
#>  [82] tm_0.7-20              miniUI_0.1.2           knitr_1.52
#>  [85] gridExtra_2.3.1        NLP_0.3-3              stats4_4.6.1
#>  [88] crul_1.6.0             xfun_0.62              graphlayouts_1.2.5
#>  [91] matrixStats_1.5.0      DT_0.34.0              humaniformat_0.6.0
#>  [94] stringi_1.8.9          qpdf_1.4.1             yaml_2.3.12
#>  [97] evaluate_1.0.5         httpcode_0.3.0         cli_3.6.6
#> [100] xtable_1.8-8           Rcpp_1.1.2             readxl_1.6.0
#> [103] triebeard_0.4.1        XML_3.99-0.25          parallel_4.6.1
#> [106] assertthat_0.2.1       pubmedR_1.0.2          slam_0.1-56
#> [109] viridisLite_0.4.3      scales_1.4.0           crayon_1.5.3
#> [112] openxlsx_4.2.9         rlang_1.3.0            fastmatch_1.1-8
```